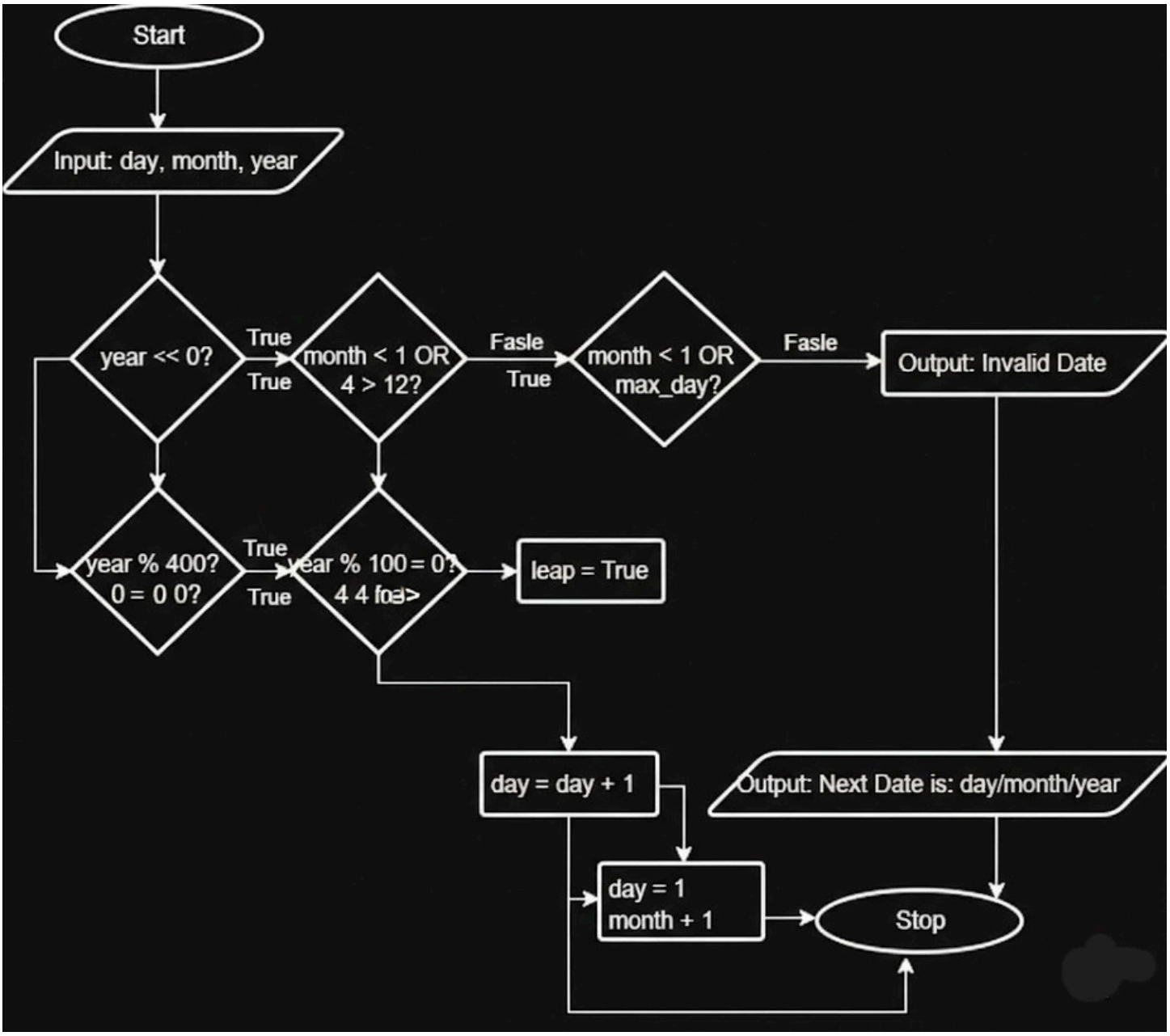
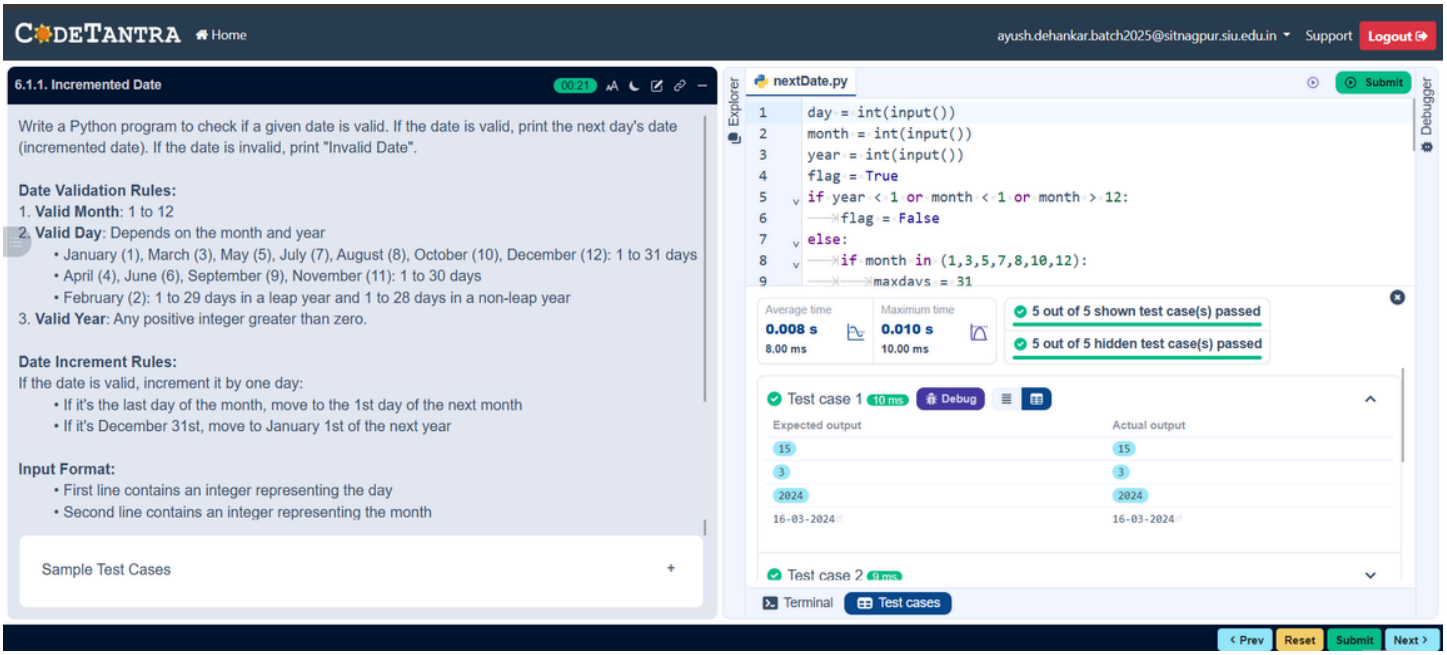
Algorithm

1. Start.
2. Read day, month, and year from the user.
3. If year ≤ 0 → Print "Invalid Date" → Stop
4. If month < 1 OR month > 12 → Print "Invalid Date" → Stop
5. If (year % 400 == 0) → Set leap = True
6. Else if (year % 100 == 0) → Set leap = False
7. Else if (year % 4 == 0) → Set leap = True
8. Else → Set leap = False
9. If month is 1, 3, 5, 7, 8, 10, or 12 → Set max_day = 31
10. Else if month is 4, 6, 9, or 11 → Set max_day = 30
11. Else if month == 2 → If leap == True → Set max_day = 29 → Else → Set max_day = 28
12. If day < 1 OR day > max_day → Print "Invalid Date" → Stop
13. Increment day by 1
14. If day > max_day → Set day = 1 → Increment month by 1
15. If month > 12 → Set month = 1 → Increment year by 1
16. Print "Next Date is: day/month/year"
17. Stop.


Start
Input: day, month, year
year << 0?
True
True
month < 1 OR
4 > 12?
Fasle
True
month < 1 OR
max_day?
Fasle
Output: Invalid Date
year % 400?
0 = 0 0?
True
True
leap = True
day = day + 1
Output: Next Date is: day/month/year
day = 1
month + 1
Stop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6.1.1. Incremented Date
00:21
Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".
Date Validation Rules:
1. Valid Month: 1 to 12
2. Valid Day: Depends on the month and year
• January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
• April (4), June (6), September (9), November (11): 1 to 30 days
• February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
3. Valid Year: Any positive integer greater than zero.
Date Increment Rules:
If the date is valid, increment it by one day:
• If it's the last day of the month, move to the 1st day of the next month
• If it's December 31st, move to January 1st of the next year
Input Format:
• First line contains an integer representing the day
• Second line contains an integer representing the month
Sample Test Cases
Explorer
nextDate.py
Submit
Debugger
day = int(input())
month = int(input())
year = int(input())
flag = True
if year < 1 or month < 1 or month > 12:
flag = False
else:
if month in (1,3,5,7,8,10,12):
maxdays = 31
Average time
0.008 s
8.00 ms
Maximum time
0.010 s
10.00 ms
5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed
Test case 1
10 ms
Debug
Expected output
Actual output
15
3
2024
16-03-2024
Test case 2
9 ms
Terminal
Test cases
Prev
Reset
Submit
Next